viStaMPS Manual

version 1.1.1

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August 2013

viStaMPS v1.1.1 was developed and tested using StaMPS v3.2.1

This version is only prepared for PS results visualization not SB and PS+SB

Next version will include PS+SB visualization features

Publications that contain figures produced by the viStaMPS software should contain an acknowledgment. [For example: This figure was generated using viStaMPS (Sousa *et al.*, 2013)].

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Contents

A. Installation/Configuration

viStaMPS can be downloaded from http://vistamps.utad.pt:

A.1. Download and execute viStaMPS.exe (32 or 64 bits)

Download the executable file and save it. viStaMPS is ready to be used. Using this modality, viStaMPS can be run without starting Matlab®.

A.2. Download viStaMPS full package

a) Download and extract `viStaMPS.rar` inside `StaMPS matlab` folder;

b) Source StaMPS_CONFIG.xxxx (see StaMPS manual), to ensure that all StaMPS and viStaMPS scripts are accessible.

> If you are using Linux, this must be done whenever a new terminal is opened. You might want to add this line to your `.cshrc` or `.bashrc` file so that this is done automatically.
>
> If you want to run viStaMPS from Matlab main window (mainly windows® users) you have to set the path to StaMPS `matlab` folder (file»Set path...)

c) Open Matlab® and execute viStaMPS.fig from guide (see Figure 1);

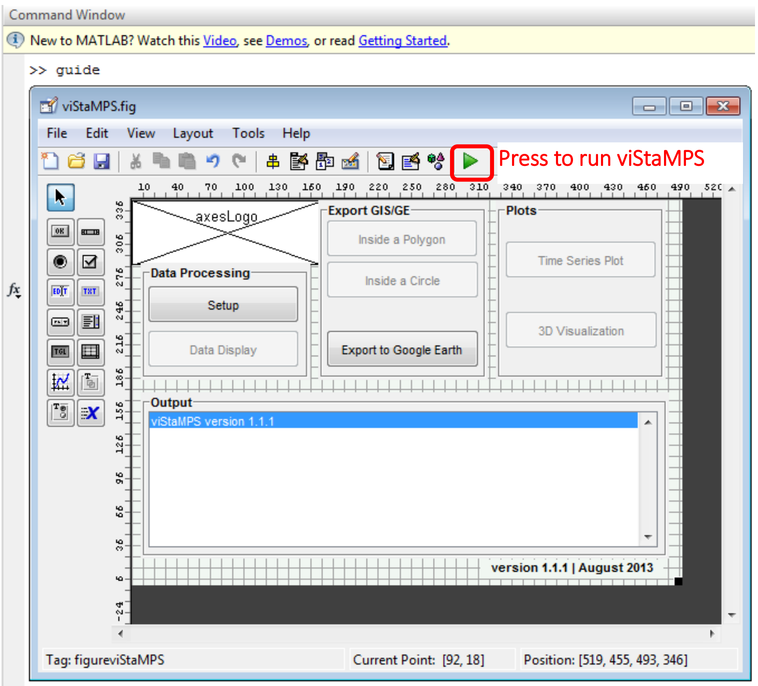

Figure 1: Open and run viStaMPS from guide.

B. Requirements

viStaMPS requires Matlab and its Image Processing Toolbox.

All the tests performed in the development of this project were done using a 2.53 GHz Intel Core™ 2 Duo processor and 4 GB of RAM memory. The computational aspects depend, of course, on the total number of PS, coherence and deformation thresholds applied, the number of images available and the parameters used for interpolation (see Sousa *et al.*, 2013 for details).

C. Files/scripts included in viStaMPS package

Script/figure name	Brief description
viStaMPS.fig	viStaMPS main figure and script. All features are controlled by these files.
viStaMPS.m	
viSSetup.fig	With these files it is possible to perform viStaMPS initial settings (set project folder, select orbit type, etc.). By running these scripts a set of files will be created in the project folder (they all have the prefix viS on their names).
viSSetup.m	
viSDataDisplay.fig	Scripts that allow the creation of deformation map (applying coherence and deformation thresholds) and phase wrapped/unwrapped figures.
viSDataDisplay.m	
viSExport.fig	Scripts that control GIS/GE exportation features.
viSExport.m	
viS_Initialize.m	Script that checks if the initial files and parameters already exists in the project folder. If not, all the necessary data will be created.
viS_cmap_defo	Colormap to be used in the deformation map.
viS_message.m	Send useful information to the user through viStaMPS output window.
viS_ps_plot.m	Adapted StaMPS `ps_plot.m` script to be run through Data Display window
viS_getcircle.m	Script to capture and save PS inside a defined circle.
viS_getpoly.m	Script to capture and save PS inside a defined polygon.
viS_getps_ts.m	Script to generate PS time series plot by picking a point in the deformation map figure.
viS_gescatter.m	Adapted StaMPS scripts used in the GE exportation.
viS_ps_gescatter.m	
viS_3DVisualization.m	Set of scripts to generate the 3D surface deformation superimposed to the mean amplitude image.
viS_smoothn.m (dctn.m ; idctn.m)	
viS_mesh.m	
viS_DataStructure.txt	viStaMPS internal data structure (useful for programmers).

1. Introduction

This manual provides a guide to running **viStaMPS**, which is made available for non-commercial applications only and can be downloaded from http://vistamps.utad.pt (see conditions of use).

Due to its proven reliability and freeware distribution among the scientific community, **Sta**nford **M**ethod for **P**ersistent **S**catterers (**StaMPS**) InSAR implementation (Hooper *et al.*,2004; Hooper *et al.*, 2007, and Hooper, 2008), which is based on the processing of multi-temporal SAR data, is widely used for ground deformation monitoring. However, some issues can make the interpretation of the results a difficult task: StaMPS supports data processing based on command prompt, which increases the difficulty of usage by users not familiar with the specificities of the programming language that supports StaMPS. Moreover, several visualization tasks are not implemented in the standard approach requiring that each user develops its own code for visualization and interpretation purposes.

viStaMPS (**vi**sual **StaMPS**) is a new visual application developed to enhance the visualization, manipulation and exportation of StaMPS results. The programmed application is developed in Matlab® through the Graphical User Interface (GUI) and no coding is required for running it, which avoids any programming language knowledge for standard uses. This tool integrates fully new features together with various scripts from StaMPS, allowing the generation of the desired plots/maps in a user-friendly interface. It is done by simply clicking some buttons or changing some parameters located in visual panels, instead of input commands in a prompt. Moreover, since it is written in Matlab, it gives the users the opportunity to access the data very easily and make their own modifications.

In summary, like StaMPS, **viStaMPS** is an open-source project and, in addition to StaMPS features, it also includes entirely new functionalities.

Being a research tool, **viStaMPS** is continually under development and will count on the dynamism of its users to improve and/or add new features.

2. Getting started with viStaMPS

Figure 2 presents the viStaMPS main window, which is divided in four main areas: **A** (Data Processing), **B** (Export GIS/GE), **C** (Plots) and **D** (output). Each operation can be accessed through different buttons and options located on the application panels. Depending on the selections made, buttons and/or parameters can toggle between active/inactive states.

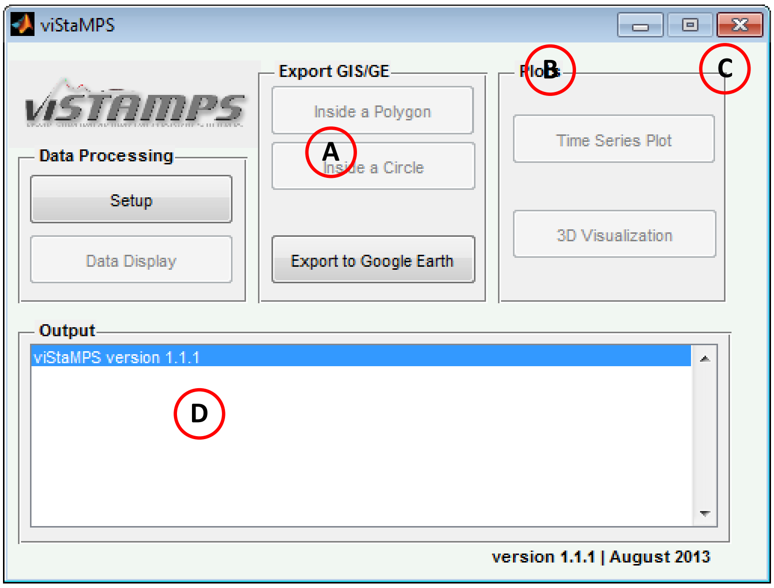

Figure 2: viStaMPS main window.

2.1. Data Processing

viStaMPS Data Processing section is composed by two buttons (`Setup` and `Data Display`). As shown in Figure 2, the first time viStaMPS is run, most of the options present inactive states. This is because the work folder is not yet defined.

2.1.1. Selecting the project folder

Before starting it is mandatory to setup the project by pressing the `Setup` button. This operation will originate the window presented in Figure 3. Follow the instructions below for project setup:

1. Browse to the desired StaMPS project folder (usually `INSAR_master_date` folder);

2. Choose the `Orbit Type` (Ascending or Descending) of the processed stack;

3. Define the `Scale Factor` (directly related to the multilook factor) and the `Distance` parameters. By default, Scale Factor is 1 (corresponding to a multilook [1 5]) and `Distance` is 5 (see later its meaning);

Scale Factor parameter plays a decisive role, since, depending on the size of your area and the memory on your computer, this value may be reduced. Generally, crops containing < 5 million SLC pixels are OK (lower values require less processing time).

The Distance parameter will be discussed in the 3D Visualization section.

4. Press Run Project button. This will create (if needed) some files that will be used in the next features (background images, LOS velocity, phase wrapped and phase unwrapped data, etc.). This information is provided in the viStaMPS Outlook area.

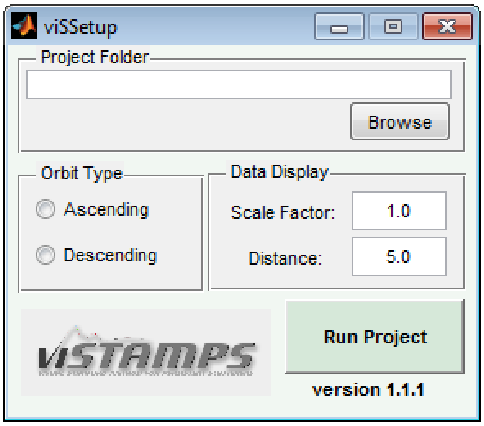

Figure 3: viStaMPS Setup window.

5. If succeeded, we are now ready to use viStaMPS features and visualize the project results.

All the files generated during this step will be saved in the project folder and have the prefix viS. Any time that we enter into the project folder again, viStaMPS checks if those files exist and if the Data Display parameters remain. If so, the creation of this setup files will be skipped.

Depending on the matlab version used, a Memory Error may occur. If so, press Run Project button again. We don't have a logic explanation for this. It seems a bug in matlab since this only happens sometimes. If the error persists, reduce Scale Factor value.

Any change in the Scale Factor value will require the creation of new project files (press Run Project button again) since all the data will be affected by this value.

6. Press Setup button (in viStaMPS main window) to close/(re)open the Setup window.

2.1.2. Data Display

After defining the project folder and confirm some needed parameters, `Data Display` button become active. Pressing this button will originate the window presented in Figure 4.

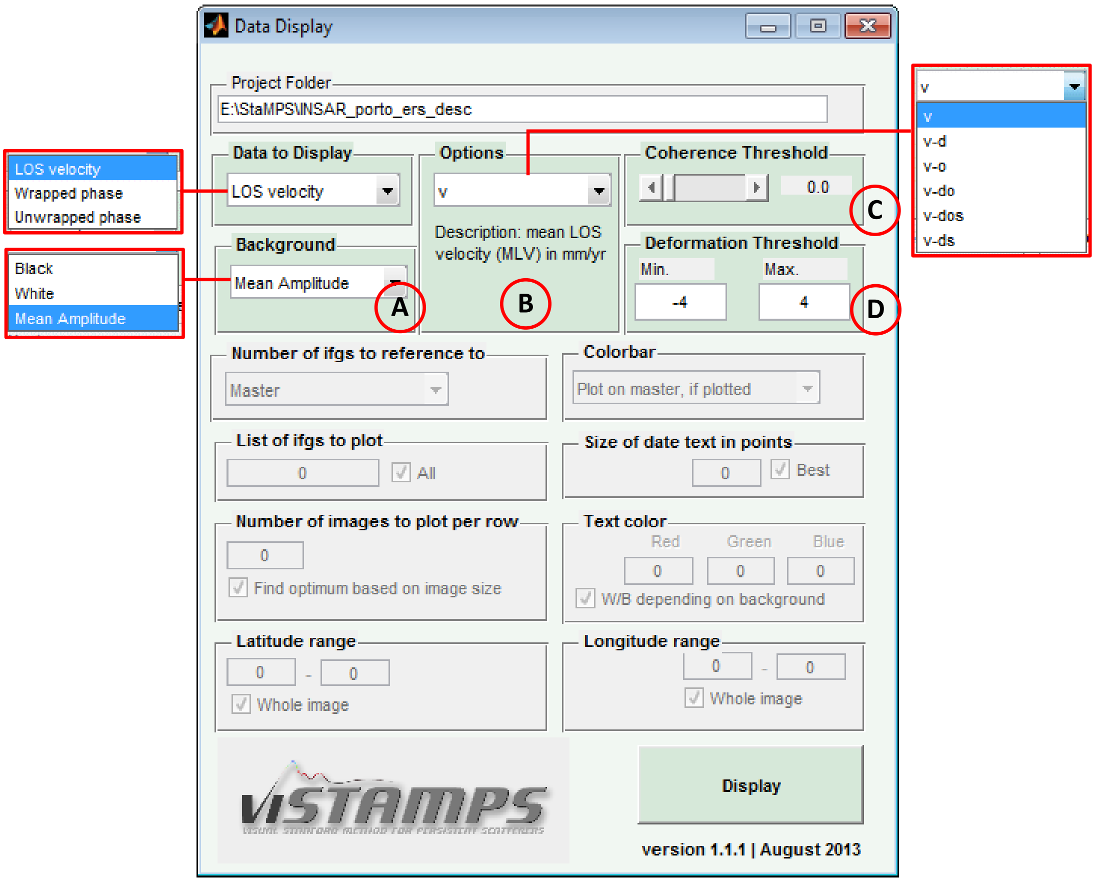

Figure 4: Data Display (default) window and available options for Data to Display, Background and Options sections.

The window presented in Figure 4 is divided in two main parts. The upper part (highlighted in blue/green) contains the parameterizations related to LOS velocity data display: (**A**) Background selection; (**B**) Value type selection (in other words, the effects of the following estimations can be removed from the displayed results: atmosphere and orbit error due to the master [option 'm'], atmosphere and orbit error due to the slave [option 's'], phase ramps [option 'o'], and DEM error [option 'd']; (**C**) Slide bar for filtering the points according to the coherence threshold selected; and (**D**) Definition of the minimum and maximum deformation rates to be displayed (by default, min and max values of the whole dataset are used).

> Some options may not be available in the value type selection (section B in Figure 4) since these values depend on the StaMPS parameters. For example, option 'a' will not be available if StaMPS parameter `scla_deramp` is set to 'n'.

After selection of the desired options, press `Display` button to get the deformation map, as shown in Figure 5.

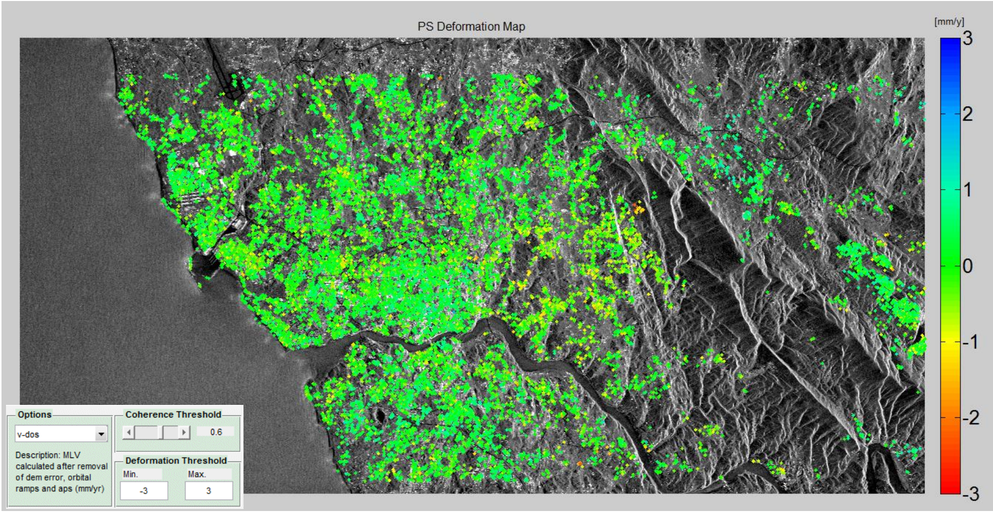

Figure 5: LOS Average velocity measured over the time period of satellite imagery analyzed for each ground measurement point identified (estimate of DEM, orbit and atmosphere influence was removed). Color scale is calculated according to the maximum and minimum data values. The mean amplitude image is used as background. Main parameters used are also shown.

Figure 6 shows some deformation maps using different thresholds of coherence and/or deformation rates.

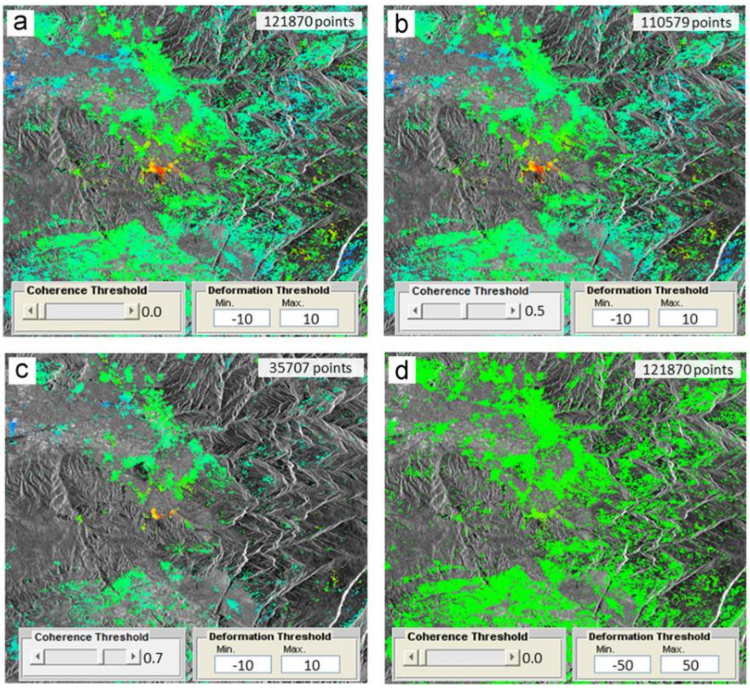

Figure 6: Examples of the influence of different thresholds for coherence and deformation in the represented PS. (a) All the points found in the processing, (b) only the points with coherence greater than 0.5, (c) only the points with coherence greater than 0.7, and (d) all the points but with different deformation thresholds bigger than the default limits.

Usually, all the points detected by StaMPS are plotted on deformation maps regardless of noise values (e.g. the coherence, a measure of the goodness of fit of the model to the observations, ranging from a minimum of 0 to a maximum of 1 (Hanssen, 2001), or the deformation rates). In comparison with the original StaMPS, viStaMPS includes the possibility of filtering the results according to deformation and coherence thresholds through a slide bar.

If the `Data Display` is set to `Wrapped phase` or `Unwrapped phase` the lower section of `Data Display` window become active. All the parameters available to run StaMPS `ps_plot.m` script (see StaMPS manual for more details) can be accessed from here. Figure 7 shows all available parameters.

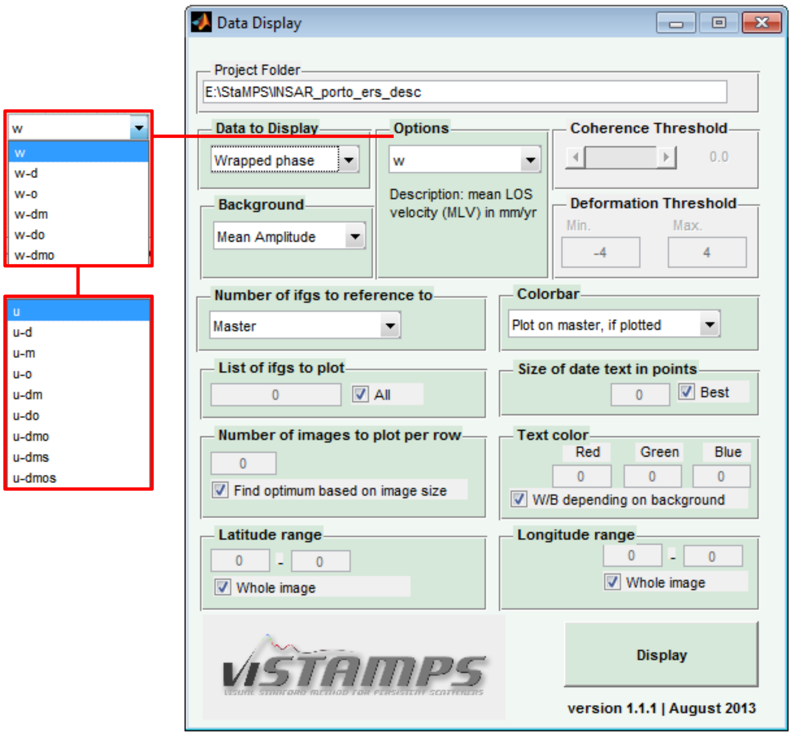

Figure 7: Data Display window for wrapped phase and available options for wrapped (w) and unwrapped (U) phase.

Figure 8 shows the unwrapped interferograms generated with the option ‘u-dmos’

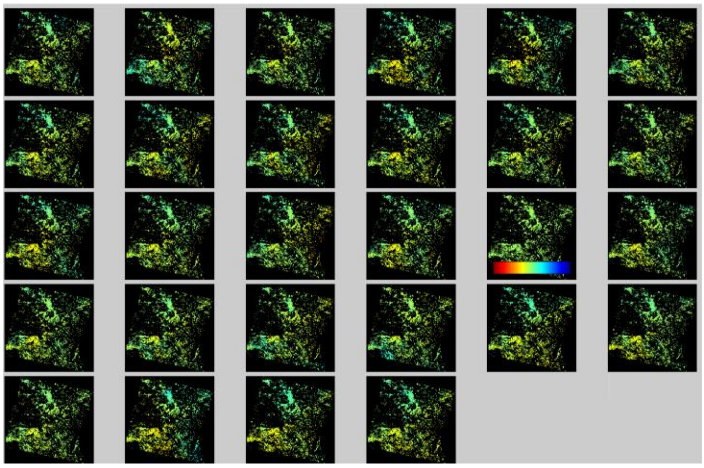

Figure 8: Unwrapped interferograms over a black background created by StaMPS ps_plot.m script included in viStaMPS.

Press `Data Display` button (in viStaMPS main window) to close/(re)open the Data Display button window.

2.2. Export GIS/GE

The last version of StaMPS (release version 3.2.1) only allows the exportation of all the points of the whole processed area by running various chained-commands/scripts. In viStaMPS, this task can be done by simply clicking a button. Detailed geocoded 2D or 3D point information can be exported to GIS (Geographical Information System) applications (e.g., Quantum GIS, ArcGIS, etc.) for example, to be used to create maps of the different data fields (average velocity, cumulative displacement, etc.) for the interpretation of ground deformation. The results can be related to the whole processed area or only to a small region defined by drawing a polygon or a circle, as shown in Figure 9. This represents a significant improvement in comparison with the original StaMPS options. An example of the structure of a '.txt' file generated by the execution of this operation is also shown in Figure 9.

This option becomes active only when a PS Velocity map (Figure 6) is open. The steps presented below must to be followed:

1. Press `Inside a Polygon` (or `Inside a Circle`) button and draw the polygon/circle, in the PS velocity map, containing the points you want to export (<u>or pick a point to define the center of the circle and a second point to define the radius</u>). All the points inside the defined polygon/circle will be selected.

> In the case of `Inside a Polygon` use <u>double click</u> or press <u>enter</u> to execute the command.
>
> In the case of `Inside a Circle` <u>click a point and drag</u> to define the circle. Move the circle and resize it using the grips. When satisfied, double click to execute the command.

2. In the window that appears, give a name to the file ('.txt') to be exported;

3. Save the colorbar image (to be used in GE).

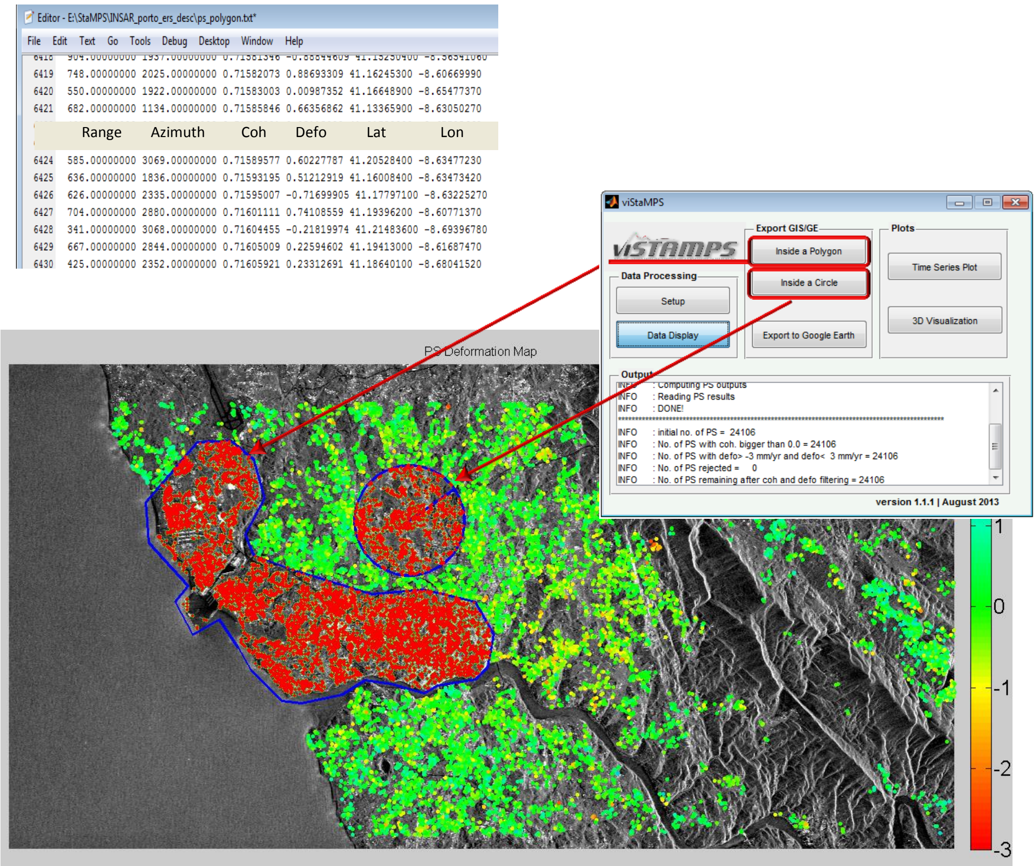

Figure 9: Example of exportation of detailed point information (range, azimuth, coherence, deformation, latitude and longitude), inside a polygon or a circle, to an external file that can be used, for instance, in a GIS software.

The buttons `Inside a Polygon` and `Inside a Circle` can be used together in the same PS deformation map allowing the exportation of different areas of the scene.

Press the button `Inside a Polygon` and draw a polygon including all the PS if you want to export all the point in the PS deformation map.

Press the button `Display` in the `Data Display` window to clean up poly/circle results drawn in the figure.

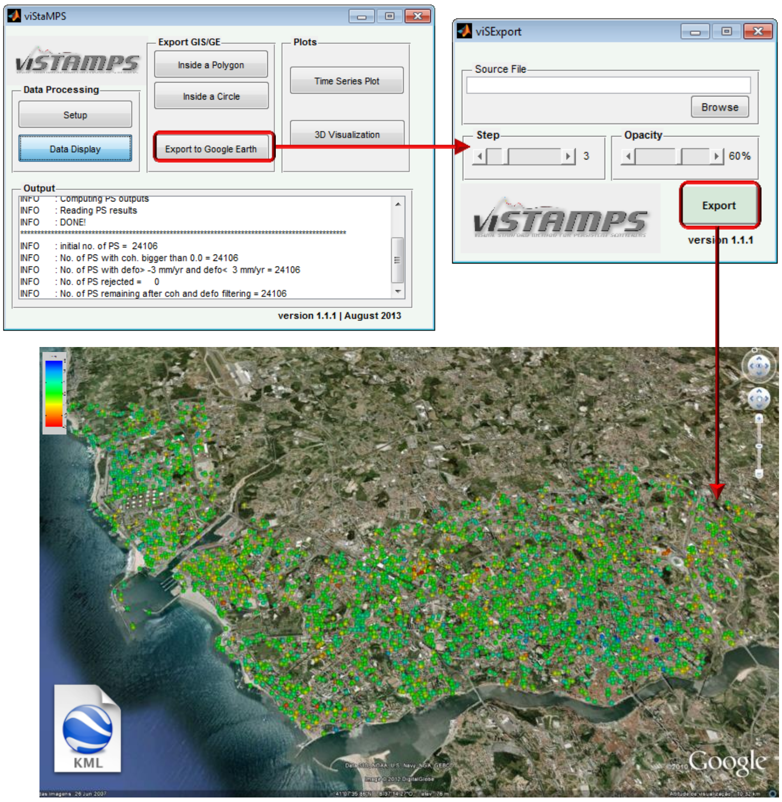

Figure 10: Example of Google Earth image with the points exported in Figure 9 using `Inside a Polygon` button. A KML file can be selected by simply clicking `Export to Google Earth` button and select the desired `Step` and `Opacity` parameters (see StaMPS manual for more details).

By pressing `Export to Google Earth` button, we will be asked to browse a file to be exported to GE '.kml' file. Choose, consecutively, one of the files exported using `Inside a Polygon` or `Inside a Circle` to generate the corresponding GE file and the desired (corresponding) colorbar image. This process and the respective result are illustrated on Figure 10. Again, this represents a significant improvement comparing with StaMPS options, because now it is possible to export any set of points according to different threshold values (deformation, coherence, etc.). While in StaMPS we are limited to export all the points in the whole processed area, in viStaMPS a smaller area can be defined together with the application of a threshold to export only the desired points.

2.3. Plots

2.3.1. Time Series Plot

Time series plot is another interesting output that helps in the analysis and interpretation of results, since it describes the evolution of displacement over the entire analyzed period. This information cannot be extracted solely from the analysis of the average velocity, cumulative displacement or average acceleration field. A displacement time series is provided for every ground measurement point identified in the StaMPS analysis, as shown in Figure 11. Each point on a time series corresponds to a single satellite acquisition being possible to identify non-linear movement, seasonal trends, ground acceleration, etc. In viStaMPS, time series plots can be visualized by simply picking a PS from the deformation map. The resulting plots will reflect the viStaMPS parameters used for plotting the deformation map.

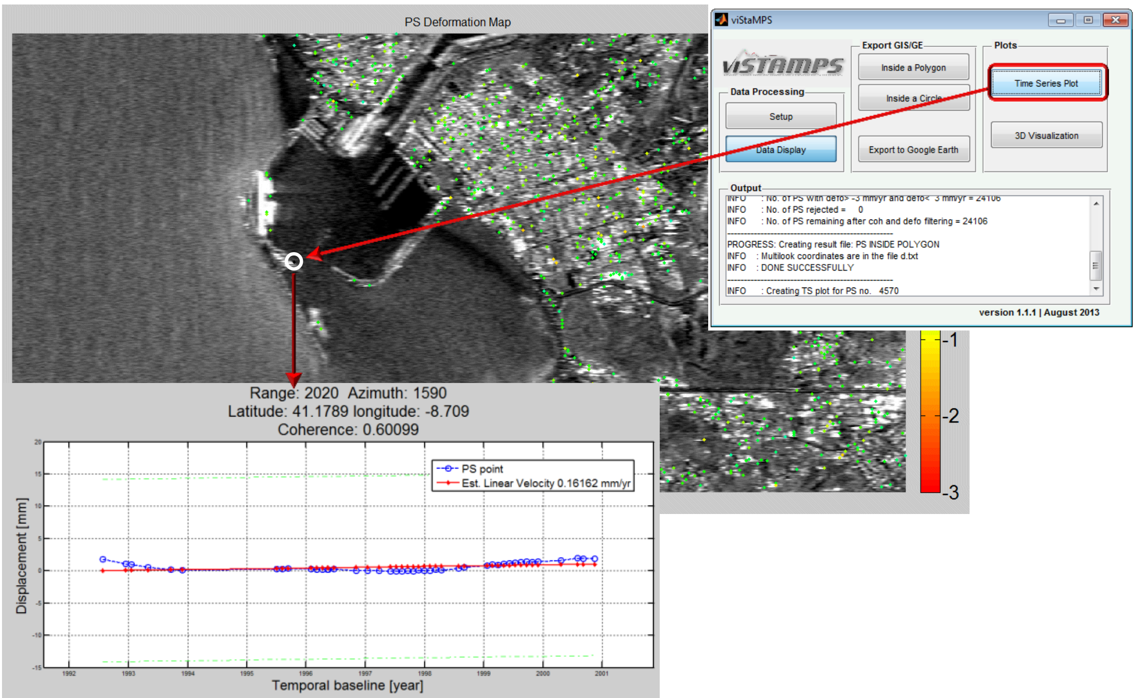

Figure 11: Example of a displacement time series plot for a single ground measurement point selected directly from the graphical display area. In this case, the selected PS, belonging to a port structure, is very stable.

2.3.2. 3D Visualization

The ability of InSAR technique to monitor wide-area ground movement at regular intervals is useful in understanding the surface response to ground instabilities and Earth activity, and provides data for the evaluation of risk models. Result from StaMPS analysis consists on discrete ground measurement points, each of which has a temporal displacement. Data can be presented as single points on a background map (previous sections), or as interpolated points creating, for instance, a 3D perspective visualization of the deformation surfaces (Figure 12). In some cases, this visualization mode can be a much more pleasant way to analyze the results.

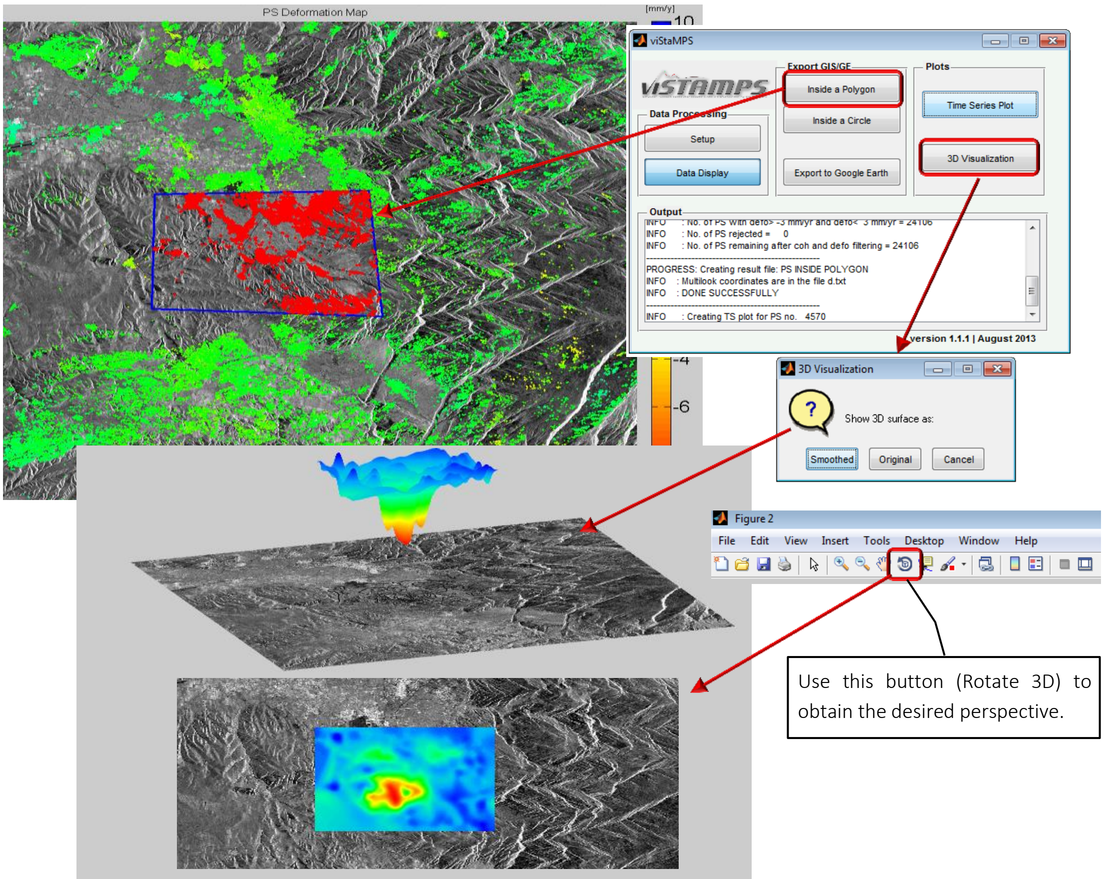

Figure 12: Example of 3D deformation plots generated by clicking the `3D Visualization` button on the viStaMPS main window. In this particular case, only the points exported using `Inside a Polygon` are used.

The `Distance` parameter shown in Figure 3 is used to generate the 3D maps presented in Figure 12. It controls the height that the 3D surface deformation is displayed above the image amplitude.

Choose `Show 3D surface as: Smoothed` if you want to remove low-frequency deformations or `Original` to show the processed deformation. Smooth factor may be modified in `viS_3DVisualization.m` script.

3. Future work

Being a research tool, viStaMPS is continually under development. Future development of viStaMPS includes the improvement of code/interface, the accomplishment of usability tests in order to help the navigation experience and, lastly, the ability to generate videos with different scenarios at the processed area projecting the deformation/consequences in the future. In a near future (next version) PS+SB visualization features will be included. viStaMPS team would like to convert, also in a near future, the full StaMPS processing chain into a visual application and include new processing techniques in order to increase the density of PSs (the Quasi Persistent Scatterers – QPS – algorithm (Perissin and Wang, 2012)will be studied and implemented).

Our final goal is to provide viStaMPS with different options for combining long series of data, and give the user the opportunity to choose which set of interferograms to process and with which techniques (PS, SB, QPS or combinations). Currently, viStaMPS is only a module dedicated to PS data visualization and exportation.

4. Conditions of use

The conditions of use are as follows:

viStaMPS is a scientific-purpose software and cannot be commercialized, nor can parts or products of it be commercialized. Our version of the software is the only official one. Please do not distribute the viStaMPS software to third parties, instead refer to the viStaMPS home page. This is in order to guarantee uniformity in the distribution of updates and information. The authors are not responsible for any damage caused by errors in the software or the documentation.

Users are very welcome to extend the capabilities of the viStaMPS software by implementing new algorithms or improving the existing ones. It is intended that if new software is developed based on viStaMPS, that this also is made available for free to the other users (through us).

We would appreciate if any addition or modification of the software would be announced first to us, so that it can be included in the official (next) version of the software.

Publications that contain figures produced by the viStaMPS software should contain an acknowledgment. [For example: This figure was generated using viStaMPS (Sousa *et al.*, 2013)].

5. Change History

5.1. Version 1.0 (June 2012)

- Initial beta release (Sousa et al., 2013);

5.2. Version 1.1 (July 2013)

- Conversion of v1.0 interface into a modular and more versatile interface;
- Addition of `Project Setup` step to improve processing speed and performance;
- Addition of `Scale Factor` and `Distance` parameters.

5.3. Version 1.1 (August 2013)

- Update to `Inside a Circle` feature;
- Optimization of `Export GIS/GE` section. Problems experienced by some users in Export PS `Inside Polygon`/`circle` fixed;
- Extra step added to Export to Google Earth in order to allow the inclusion of the colorbar in GE visualization.

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